

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VII

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Class: TYIT	Batch: 1
Date of Assignment: 4/8/26	Date/Time of Submission: 4/8/26

AIM:

a. Develop a PWA that works in offline mode using Service Worker.

CODE:

offline.html:

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>College Portal</title>
  <style>
    body {
      margin: 0;
      font-family: Arial,sans-serif;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      background: #f8f9fa;
      color: #333;
    }

    .box {
      text-align: center;
      padding: 30px;
    }

    h2 {
      margin: 10px 0;
    }

    p {
      color: #666;
    }

    button {
      padding: 10px 20px;
      border: none;
      border-radius: 5px;
      background: #0d6efd;
```

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```
        color: #fff;
        cursor: pointer;
    }
</style>
</head>
<body>
    <div class="box">
        <h2> Cat Portal</h2>
        <p>All cats server are busy please try again.</p>
        <button onclick="location.reload()">Try Again</button>
    </div>
</body>
</html>
```

manifest.json:

```
{
  "name": "College Information Portal",
  "short_name": "CIP",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "icon/ani.png",
      "sizes": "128x128",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "icon/animal.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

service-worker.js:

```
self.addEventListener("install", e => {
  e.waitUntil(
    caches.open("v1").then(c =>

      c.add("/offline.html"))
  );
});

self.addEventListener("fetch", e => {
  e.respondWith(
    fetch(e.request).catch(() =>
      caches.match("/offline.html"))
  );
});
```

_Layout.cshtml:

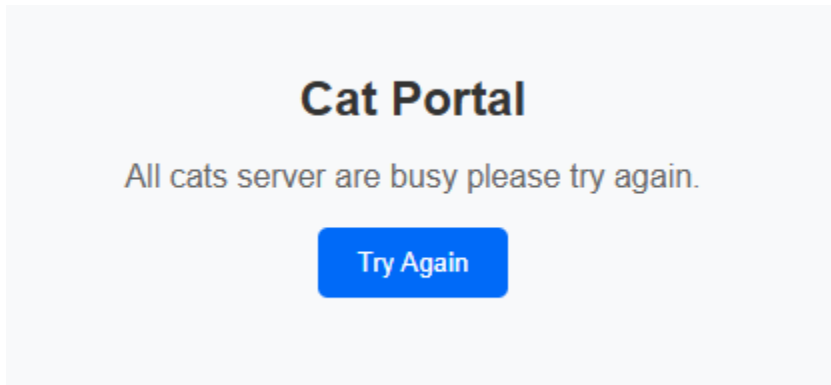
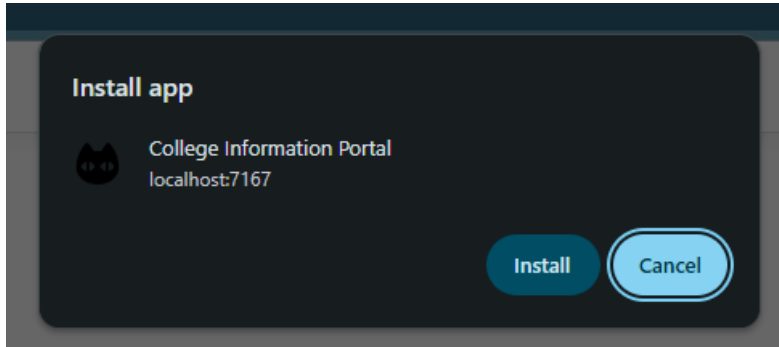
Before ending head tag

```
<link rel="manifest" href="manifest.json"/>
```

Before ending body tag

```
<script>
if ('serviceWorker' in navigator)
{
    navigator.serviceWorker.register('/service-worker.js');
}
</script>
```

OUTPUT:



b. Create a Student Record Management PWA.

CODE:

Index.cshtml:

```
@{
    ViewData["Title"] = "Home Page";
}

<div class="text-center">
    <h1 class="display-4">Welcome</h1>
    <p>Learn about <a href="https://learn.microsoft.com/aspnet/core">building Web apps with ASP.NET Core</a>.</p>
    <h1>Offline Page</h1>
    <h2>Student Record</h2>

    <p>Roll No : @Model.RollNo</p>

    <p>Name : @Model.Name</p>
```

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</div>

HomeController.cs:

```
using Microsoft.AspNetCore.Mvc;
using Prac6Practice.Models;
using practical7.Models;
using System.Diagnostics;
namespace Prac6Practice.Controllers
{
    public class HomeController : Controller
    {
        public IActionResult Index()
        {
            Student s = new Student();

            s.RollNo = 39;
            s.Name = "Bhavesh";

            return View(s);
        }

        public IActionResult Privacy()
        {
            return View();
        }

        [ResponseCache(Duration = 0, Location = ResponseCacheLocation.None, NoStore = true)]
        public IActionResult Error()
        {
            return View(new ErrorViewModel { RequestId = Activity.Current?.Id ?? HttpContext.TraceIdentifier });
        }
    }
}
```

Student.cs:

```
namespace Prac6Practice.Models
{
    public class Student
    {
        public int RollNo { get; set; }
        public string Name { get; set; }
    }
}
```

service-worker.js:

```
const CACHE = "v1";

self.addEventListener("install", e => {
    e.waitUntil(
        caches.open(CACHE).then(cache => {
            return cache.addAll([
                "/",
                "/Home/Index",
            ])
        })
    )
})
```

```
        "/css/site.css",  
        "/js/site.js"  
    });  
    })  
    );  
});  
  
self.addEventListener("fetch", e => {  
    e.respondWith(  
        caches.match(e.request).then(response => {  
            return response || fetch(e.request);  
        })  
    );  
});
```

OUTPUT

Welcome

Learn about [building Web apps with ASP.NET Core](#).

Offline Page Student Record

Roll No : 39

Name : Bhavesh